Henok Mesfin

Software Engineering Student · Python & JavaScript · Full-Stack Developer · Addis Ababa

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Addis Ababa, Ethiopia (UTC+3)
Third-year BSc Software Engineering
Open to remote contract work

PROFILE

Third-year software engineering student with three years of building across Python and JavaScript. I write and review code against precise specifications — Ethiopia's ERCA tax schedule as a payroll engine, tenant isolation with row-level security, and a 20-scenario behavioural test suite for AI assistants that runs on every commit. Comfortable debugging unfamiliar systems, designing REST APIs, and giving exacting, evidence-backed technical feedback.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, HTML/CSS (TypeScript in progress)
Backend & APIs: FastAPI, Next.js route handlers, REST API design, SQLAlchemy 2.0, Mongoose, JWT/OAuth2, RBAC
Data & systems: PostgreSQL, MongoDB, Supabase, SQLite, schema design, indexing, Row Level Security, tenant isolation
Frontend: React 19, Next.js 15 (App Router), Tailwind CSS, responsive UI
Testing & QA: Behavioural test design, prompt-injection and hallucination evaluation, CI regression suites
Tooling: Git/GitHub, Docker & Compose, GitHub Actions, Vercel, Linux/CLI, technical documentation

TECHNICAL EXPERIENCE

EthioPayroll — Full-Stack Developer

Self-directed, open source
2026 — Present

Payroll, HR and compliance SaaS for Ethiopian businesses

- Implemented Ethiopia's ERCA payroll engine — progressive tax across six brackets, a 7% / 11% pension split, four overtime multipliers — as one shared module, verified bracket-by-bracket against the published schedule.
- Designed a multi-tenant schema (12 models, 21 REST routes) with role-based access control, scoping every query by tenant so no request can cross a company boundary.
- Diagnosed a duplicate-query and duplicate-subscription pattern that fired on every dashboard navigation, and fixed it by lifting profile resolution into one React context provider.

Next.js 15 · React 19 · MongoDB · Mongoose · NextAuth · Tailwind CSS · Recharts

Backend, Security & AI Quality — Backend Developer

Self-directed, open source
2025 — 2026

Authenticated APIs, hardened deployments, behavioural evaluation of AI assistants

- Built a JWT-authenticated CRUD API in FastAPI with bcrypt hashing and ownership checks in every query — 47 commits, one codebase running locally, in Docker, and serverless on Vercel.
- Authored a 20-scenario behavioural suite for AI assistants — hallucination, prompt injection, privacy leaks, ambiguity, unauthorised actions — run in CI on every push.
- Reviewed a 16/20 first pass, isolated four failure modes, and documented each fix in a written case study with a reusable evaluation rubric.

Python · FastAPI · SQLAlchemy 2.0 · PostgreSQL · Docker · GitHub Actions · Vercel

Full-Stack Foundations — Self-taught Developer

Self-directed
2023 — 2025

- Learned Python and JavaScript from first principles — CLI tools with structured exception handling, framework-free HTML/CSS layouts, then React, async data fetching and relational schemas.

Python · JavaScript · HTML & CSS · React · Git

SELECTED PROJECTS

EthioPayroll — Multi-tenant payroll and HR SaaS: 12 models, 21 REST routes, 25 RLS policies. Next.js · MongoDB · NextAuth

Task Management REST API — JWT-authenticated FastAPI service; one codebase, local, Docker and serverless. Python · FastAPI · PostgreSQL · Docker

AgentProof Starter — Behavioural QA kit for AI assistants: 20 scenarios in CI, published case study and rubric. Python · YAML · GitHub Actions

Pediatric Dose Reference — Offline clinical calculator: 116 drugs, 164 dosing lines, weight-based dosing. JavaScript · HTML · CSS

EDUCATION

HiLCoE School of Computer Science & Technology

Addis Ababa, Ethiopia

BSc Software Engineering, third year. Coursework: data structures and algorithms, databases, web development.

Open source & self-directed work

2023 — Present

Eight public repositories, each documented and deployed where it can be — not tutorial forks.

Amharic (native) · English (professional) · 8 public repositories at github.com/henok-ne · Code samples on request.